SULIT

UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER II , 2016 / 2017

SUBJECT CODE	:	**SCSJ 2013**
SUBJECT NAME	:	**DATA STRUCTURE AND ALGORITHM**
SECTION	:	**ALL**
TIME	:	**2.30 PM – 5.00 PM**
DATE/DAY	:	**13 JUNE 2017 (TUESDAY)**
VENUES	:	**BK 2**

INSTRUCTIONS :

This test book consists of two parts (2) parts. Answer **ALL QUESTIONS.**
The layout of the questions are as follows:

PART A	-	**True/ False Questions**	**(10 Questions : 20 marks)**
PART B	-	**Structured Questions**	**(5 Questions : 80 marks)**

ANSWER ALL QUESTIONS IN THE ANSWER BOOKLET PROVIDED.

(Please Write Your Lecture Name And Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This questions paper consists of **TWELVE (12)** printed pages including this page.

PART A : TRUE/FALSE **[20 marks]**

There are **10 Questions** in this section. Determine whether the statements given are **TRUE** or **FALSE**. Give justification for your answer.
(1 mark for TRUE/FALSE answer and 1 mark for the justification.)

1. The best case for algorithm efficiency in a sequential search is when the search key is at index 0. **TRUE/ FALSE**

   Justification

---

2. Binary search is more suitable on large data set as the search key is compared with middle value of the list iteratively to remove as much data as possible. **TRUE/ FALSE**

   Justification

Figure 1 shows 3 nodes and the available pointers. Answer question 3 and 4 based on Figure 1.

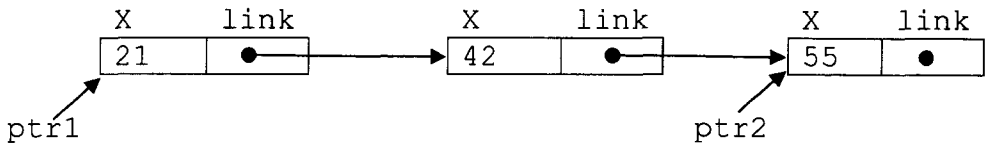


Figure 1

3. The correct statement to change **`ptr1`** to point to the next node is : **TRUE/ FALSE**

   ```
   ptr1 = ptr1->next
   ```

   Justification

---

4. The boolean expression: (**`ptr2==NULL`**) will return true, because `ptr2` is currently points to the tail node (last node in the list). **TRUE/ FALSE**

   Justification

---

5. Stack data structure implement **FIRST IN FIRST OUT (FIFO)** concept. **TRUE/ FALSE**

Justification

---

6. For array implementation of stack, the following statements must be used in order to push an item into the stack. **TRUE/ FALSE**

```
item = stack[top];
top = top - 1;
```

Justification

---

7. Both queue implementation linear linked list and circular linked list require two pointers to access the front and back of the queue. **TRUE/ FALSE**

Justification

---

8. The purpose of the source codes below is to insert an item into a circular array queue. **TRUE/ FALSE**

```
front = (front + 1 % MAX_QUEUE)
count --;
```

Justification

---

9. If the following numbers are inserted into an empty binary search tree in the following order: `10, 1, 3, 5, 15, 12, 16`. The binary search tree generated is a balanced tree. **TRUE/ FALSE**

Justification

---

10. The worst case time complexity for search, insert and delete operations in a Binary Search Tree is **O(n).** **TRUE/ FALSE**

Justification

PART B

THIS PART CONSISTS OF THREE (5) QUESTIONS. ANSWER ALL QUESTIONS.

Question 1 **[12 marks]**

Given **DATA** array of integers as in Figure 2, answer question 1 (a) and 1 (b).

index	[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
	250	270	111	135	98	100	125	170	190

Figure 2: DATA array

a) Name the searching technique suitable to be performed on the data array in Figure 1. Give the reason to justify your answer.

[2 marks]

b) What should be done to the **DATA** array in Figure 1 to reduce the complexity time of the searching process? Explain your answer.

[2 marks]

Given **NUM** array of integers as in Figure 3, answer question 1(c), 1(d) and 1(e).

index	[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
	1	3	5	7	9	11	13	17	19

Figure 3: NUM array

c) Perform Binary search for searching **key 20** on the **NUM** array in Figure 3. Show the tracing of your search using variables **`left`**, **`right`**, **`middle`**, **`NUM[middle]`** and **`found`** as shown in the table format below.

[3 marks]

`left`	`right`	`middle`	`NUM[middle]`	`found`

d) Perform Binary search for searching **key 5** on the **NUM** array in Figure 3. Show the tracing of your search using variables **`left`**, **`right`**, **`middle`**, **`NUM[middle]`** and **`found`** as shown in the table format below.

[3 marks]

left	right	middle	NUM[middle]	found

e) Discuss the differences between the number of steps and the complexity time when searching for **key equals to 19** on **NUM** array in Figure 2 using binary search and sequential search technique.

[2 marks]

Question 2 **[19 marks]**

a) A queue can be implemented using circular array or circular linked list data structure. Give one advantage of queue using circular array and one advantage of queue using circular linked list. What kind of problem that both implementation can solve.

[2 marks]

b) Give the differences between the implementation of the queue implementation array and queue implementation circular array in the following operations:

i) Queue declaration

ii) **`isEmpty()`** function.

iii) **`isFull()`** function.

[3 marks]

c) Figure 4 shows the current content of a circular queue.

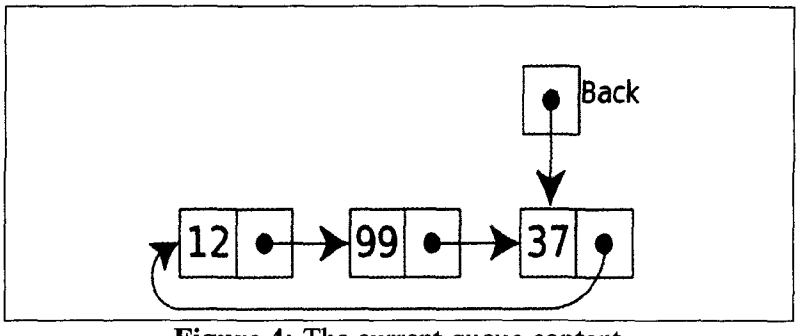


Figure 4: The current queue content

Redraw the queue in Figure 4 that shows the current content of the queue after the sequence of instructions in Figure 5 is executed.

[3 marks]

```
Queue -> enQ(9);
Queue -> enQ(4);
Queue -> deQ();
```

Figure 5 : Queue instructions

Answer the following questions, which deal with operations on queue implementation array. Figure 6 shows Queue 1 with maximum content of 4 items. Currently, there are two items already in the queue after **enQueue (A)** and **enQueue (B)** operations have been executed.

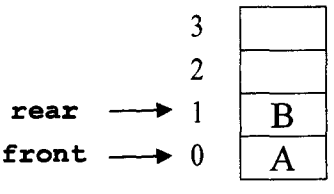


Figure 6: Queue 1

d) Perform the following tasks to Queue 1:

i. Redraw Queue 1 after **enQueue(G)** followed by **enQueue(F)**. Label the correct location of the **front** and **rear**.

ii. Redraw the queue in question (d- i) after **deQueue()** and **enQueue(G)**. Label the correct location of the **front** and **rear**.

[3 marks]

e) Figure 7, shows a circular queue called Queue 2, which fits maximum of 8 items. The current content of Queue 2 are as shown in the figure.

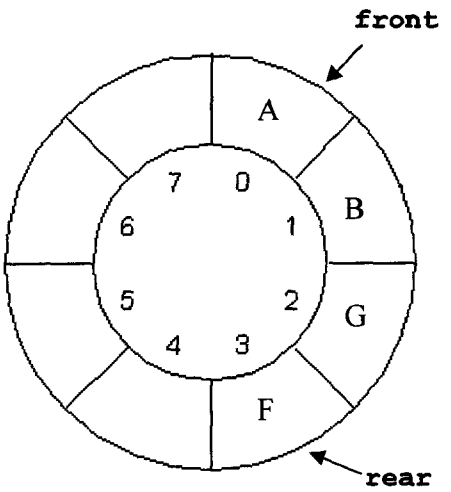


Figure 7: Queue 2

i. Redraw Queue 2 after **E, J, M, H** are **enQueue** in sequence. Label the correct location of the **`front`** and **`rear`**.

[2 marks]

ii. Redraw updated queue in question (i) after **3 deQueue** operations are performed . Label the correct location of the **`front`** and **`rear`**.

[2 marks]

iii. List all the deQueue items in question (ii) in FIFO sequence.

[1.5 marks]

iv. Redraw updated queue in (ii) after **R and Q** are **enQueue** in sequence. Label the correct location of the **`front`** and **`rear`**.

[1 marks]

v. After a series of **enQueue** and **deQueue** operations, compare the efficiency of memory usage for Queue1 and Queue2.

[1.5 marks]

Question 3 **[19 marks]**

a) Discuss the advantage of full tree and complete tree in terms of complexity related to insertion, deletion and searching operations. [2 marks]

b) Discuss the drawbacks of Binary Search Tree in terms of complexity related to insertion, deletion and searching operations. [2 marks]

c) Draw the expression tree for the following arithmetic expression. [3 marks]

a * b – (c * 11 / d) + (e + f * 5)

d) Give the inorder, preorder and postorder traversal of the tree in Figure 8. [6 marks]

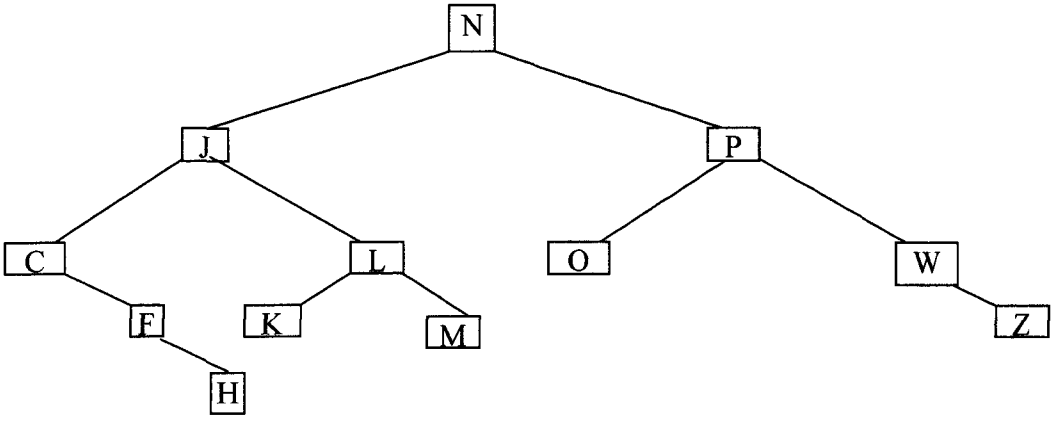


Figure 8: Binary Search Tree of char value

Given the following binary search tree in Figure 9, answer question 5 and 6.

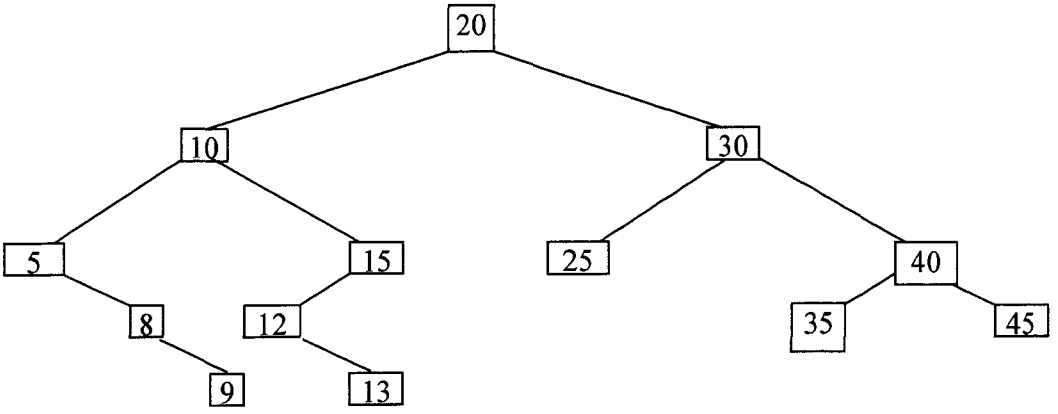


Figure 9: Binary Search Tree of int value

e) Redraw the tree in Figure 9 after nodes with values **18, 16, 19, 33 and 43** are inserted in sequence. [2 marks]

f) Redraw the tree in Figure 9 after the following nodes are deleted **in sequence**. Show the new tree after every deletion.

i) 25

ii) 20

iii) 10

[4 marks]

Question 4 **[15 marks]**

You are given the following code as shown in Figure 10. When the code execute, the result can be represented in Figure 11.

```
struct NODE {
    int item;
    NODE* prev;
    NODE* next;
};

// main function
int main(int argc, char *argv[]) {
    NODE* n1 = new NODE;
    n1->item = 6;
    n1->prev = NULL;
    n1->next = NULL;

    //assume the additional code segment for question a - d is here

}//end main
```

Figure 10 : Linked List Codes

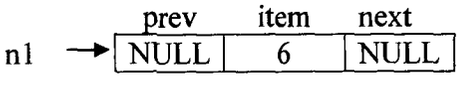


Figure 11: Current execution state with node n1

a) Show in a diagram, how nodes **n1, n2** and **n3** are interconnected to each other, when the following code segment is executed.

[3 marks]

```
NODE* n2 = new NODE;
n2->item = 3;
n2->prev = n1;
n2->next = NULL;

NODE* n3 = new NODE;
n3->item = 8;
n3->prev = n2;
n3->next = NULL;
```

b) Show in a diagram, how nodes **n1, n2** and **n3** are interconnected to each other, when the following code segment is executed.

[3 marks]

```
n1->next = n2;
n2->next = n3;
```

c) A new node, **n4** is created as follows:

```
NODE* n4 = new NODE;
n4->item = 7;
n4->prev = NULL;
n4->next = NULL;
```

Show in a diagram, how nodes **n1, n2, n3** and **n4** are interconnected to each other, when the code is executed

[3 marks]

d) Write a code segment to delete node **n2** and replace it with the node n4

[3 marks]

e) Another code segment is added as follows:

```
NODE * print_ptr;
print_ptr = n1;
```

Write a code segment to display the value of item in all the nodes, using **print_ptr**

[3 marks]

Question 5 **[15 marks]**

a) Consider the following source codes in Program Segment 1 that will be implemented on stack using array. The **size of the stack is 5.**

```
1   Stack s;
2   string a1, a2, a3;
3   :
4   :
5   a1=s.stackTop( );
6   a2= "Luck";
7   s.push(a2);
8   s.push(a1+a2);
9   s.push("Congratulations");
10  a2=s.stackTop( );
11  s.pop();
12  s.pop();
13  s.push("Bye!");
14  a3=s.stackTop( );
15  s.pop();
16  cout<< a1 << a3 <<endl;
17  cout<< a2 <<endl;
```

Program Segment 1: Stack source codes

i) Show the content of the stack after each implementation of the sources codes based on the situation below. Assume the current content of the stack before statement Line 5 is executed as shown in first stack in the Table 2. [3 marks]

Element of array	Content of Stack						
[4]							
[3]							
[2]	**Good**						
[1]	**Bye!**						
[0]	**Thank You**						
	Current content before Line 5	Content After Line 5-8 are executed	Content After Line 9	Content After Line 11	Content After Line 12	Content After Line 13	Content After Line 15

Table 2

ii) What will be displayed by the program in Program Segment 1? [2 mark]

b) Evaluate the following postfix expression using stack concept: **14 2 / 6 – 10 + 5 *** You need to use the format given in the table below in order to show the evaluation process. [5 marks]

Postfix	Ch	Opr	Opn1	Opn2	Result	Stack

b) Given the declaration of class **Stack** in Program Segment 2, answer the following question.

```
class Stack
{
    private:
        int top;
        char data[MAXSTACK];

    public:
        Stack();
        bool isFull();
        bool isEmpty();
        void push(char);
        char pop();
        void printLIFO();
        void printPop()
};
```

Program Segment 2: Declaration of class Stack

Complete the code for **pop()** function so that it will return the popped item (**char**) if the stack not empty and return an asterisk '*' if the stack is empty.

[5 Marks]